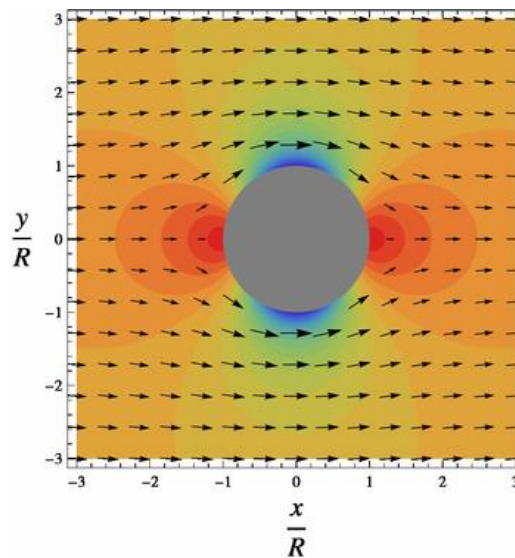


Summary: Non-lifting flow over a cylinder

- Attached flow.
- No viscous effect, No Boundary Layer.
- Flow slips on the surface.
- No wake (behind the model) – Since there is no flow separation.
- The inviscid irrotational, incompressible flow over a circular cylinder of radius R can be synthesized by adding a uniform flow with velocity U_∞ and a doublet of strength k .
- R is related to U_∞ and k through $R = \sqrt{k/2\pi U_\infty}$.
- The dividing streamline is defined as, $\psi = U_\infty r \sin\theta \left(1 - \frac{R^2}{r^2}\right) = 0$ (it goes through the stagnation points).
- The flow inside $\Psi = 0$ comes from the doublet and the flow outside $\Psi = 0$ comes from the uniform flow.
- Therefore, we can replace the flow inside the circle with a solid body, and the external flow will not know the difference.



Flow around a stationary cylinder (Source: Wikipedia)

- From the figure, we can see that the pressure distribution is symmetrical about both axes.
- As a result, the pressure distribution over the top of the cylinder is exactly balanced by the pressure distribution over the bottom of the cylinder (i.e., **there is no net lift**).
- Similarly, the pressure distribution over the front of the cylinder is exactly balanced by the pressure distribution over the back of the cylinder (i.e., **there is no net drag – D'Alembert's Paradox**).

- The velocity distribution on the surface of the cylinder is given by ($r = R$)

$$u_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (U_\infty r \cos \theta) \left(1 - \frac{R^2}{r^2} \right) = \left(1 - \frac{R^2}{r^2} \right) U_\infty \cos \theta = 0$$

$$u_\theta = -\frac{\partial \psi}{\partial r} = - \left[(U_\infty r \sin \theta) \frac{2R^2}{r^3} + \left(1 - \frac{R^2}{r^2} \right) (U_\infty \sin \theta) \right] = - \left(1 + \frac{R^2}{r^2} \right) U_\infty \sin \theta = -2U_\infty \sin \theta$$

When $r = R$,

$$U_r = 0$$

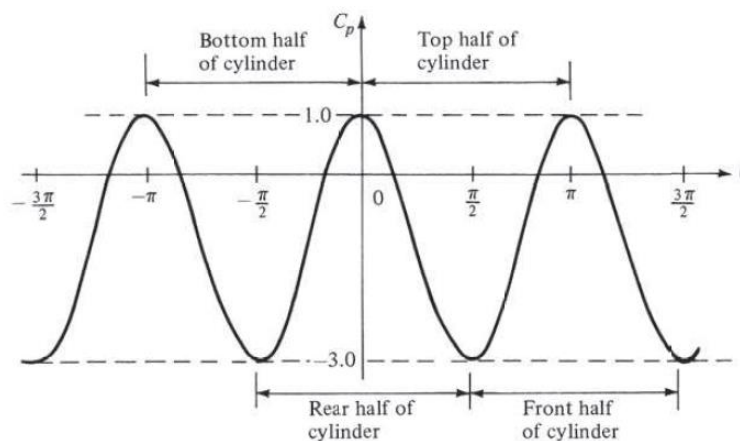
$$U_\theta = -2U_\infty \sin \theta$$

- U_θ is positive in the direction of increasing θ , that is, in the counter-clockwise direction.
- Negative sign indicates that the surface velocity for $0 \leq \theta \leq \pi$ is in the opposite direction of increasing θ .
- For $\pi \leq \theta \leq 2\pi$, the surface velocity is in the same direction of increasing θ .
- The pressure coefficient is given by

$$C_p = 1 - (V/V_\infty)^2$$

- For a cylinder, $V = U_\theta = -2U_\infty \sin \theta$ (**Note:** $U_\infty = V_\infty$)
- Hence, the pressure coefficient can be written as

$$C_p = 1 - (V/V_\infty)^2 = 1 - (-2V_\infty \sin \theta / V_\infty)^2 = 1 - 4 \sin^2 \theta$$



Pressure coefficient distribution over the surface of a circular cylinder; theoretical results for inviscid, incompressible flow.

Summary: Flow over a rotating cylinder

The velocity on the surface of the cylinder ($r = R$) is given by

$$u_r = \frac{1}{r} \frac{\partial \psi}{\partial \theta} = \frac{1}{r} (U_\infty r \cos \theta) \left(1 - \frac{R^2}{r^2} \right) = \left(1 - \frac{R^2}{r^2} \right) U_\infty \cos \theta$$

$$u_\theta = V = -\frac{\partial \psi}{\partial r} = -\left(1 + \frac{R^2}{r^2} \right) U_\infty \sin \theta - \frac{\Gamma}{2\pi r}$$

In turn, the pressure coefficient is defined as,

$$C_p = 1 - (V/V_\infty)^2 = 1 - \left(-\left(1 + \frac{R^2}{r^2} \right) U_\infty \sin \theta - \frac{\Gamma}{2\pi r} \right)^2 / U_\infty^2$$

$$C_p = 1 - \left[4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi R V_\infty} + \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2 \right]$$

The aerodynamic force coefficients can be obtained by integrating the pressure coefficient and skin friction coefficient over the surface. For inviscid flow, $\tau = 0$.

$$C_d = \frac{1}{c} \int_{LE}^{TE} (C_{p,u} - C_{p,l}) dy = \frac{1}{c} \int_{LE}^{TE} C_{p,u} dy - \frac{1}{c} \int_{LE}^{TE} C_{p,l} dy$$

In the polar coordinates,

$$y = R \sin \theta \Rightarrow dy = R \cos \theta d\theta$$

For a cylinder, $c = 2R = \text{Diameter}$. Hence,

$$C_d = \frac{1}{2} \int_{\pi}^0 C_{p,u} \cos \theta d\theta - \frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \cos \theta d\theta$$

- $\theta \Rightarrow 0 \rightarrow \pi$ represents the top surface of the cylinder
- $\theta \Rightarrow \pi \rightarrow 2\pi$ represents the bottom surface of the cylinder

$$C_d = -\frac{1}{2} \int_0^{2\pi} C_p \cos \theta d\theta$$

Substitute $C_p = 1 - \left[4 \sin^2 \theta + \frac{2\Gamma \sin \theta}{\pi R V_\infty} + \left(\frac{\Gamma}{2\pi R V_\infty} \right)^2 \right]$ into the C_d equation, we get

$$C_d = 0$$

We know that,

$$\int_0^{2\pi} \cos \theta d\theta = 0$$

$$\int_0^{2\pi} \sin^2 \theta \cos \theta d\theta = 0$$

$$\int_0^{2\pi} \sin \theta \cos \theta d\theta = 0$$

The drag on a cylinder in an inviscid, incompressible flow is zero, regardless of whether or not the flow has circulation about the cylinder.

The lift on the cylinder can be evaluated in a similar manner as follows.

$$C_l = \frac{1}{c} \int_{LE}^{TE} (C_{p,l} - C_{p,u}) dy = \frac{1}{c} \int_{LE}^{TE} C_{p,l} dy - \frac{1}{c} \int_{LE}^{TE} C_{p,u} dy$$

In the polar coordinates,

$$x = R \cos \theta \Rightarrow dx = -R \sin \theta d\theta$$

So

$$C_l = -\frac{1}{2} \int_{\pi}^{2\pi} C_{p,l} \sin \theta d\theta + \frac{1}{2} \int_{\pi}^0 C_{p,u} \sin \theta d\theta$$

$$C_l = -\frac{1}{2} \int_0^{2\pi} C_p \sin \theta d\theta$$

We know that,

$$\int_0^{2\pi} \sin \theta d\theta = 0; \int_0^{2\pi} \sin^3 \theta d\theta = 0; \int_0^{2\pi} \sin^2 \theta d\theta = \pi$$

We obtain

$$C_l = \frac{\Gamma}{RV_{\infty}}$$

From the definition, the lift per unit span can be obtained from

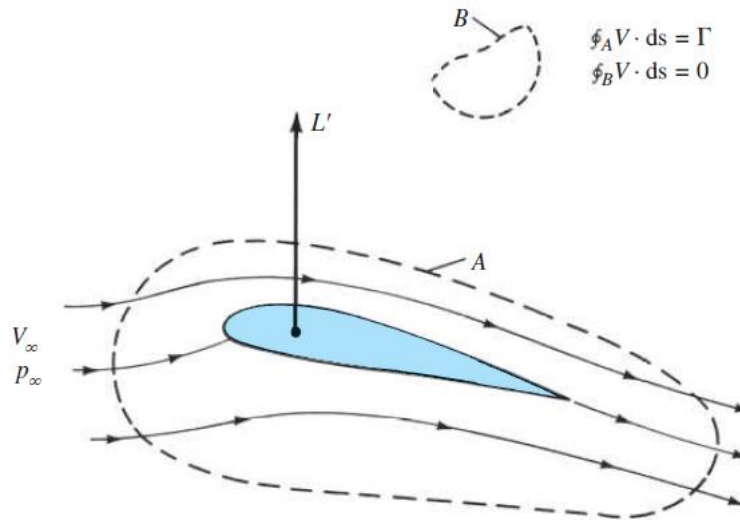
$$L' = QSC_l = 0.5\rho V_{\infty}^2 SC_l = 0.5\rho V_{\infty}^2 2R(\Gamma/RV_{\infty})$$

$$L' = \rho V_{\infty} \Gamma$$

Kutta - Joukowski Theorem states that *the lift per unit span is directly proportional to circulation.*

Let curve A be any curve in the flow *enclosing* the airfoil. If the airfoil is producing lift, the velocity field around the airfoil will be such that the line integral of velocity around A will be finite, that is, the circulation is also finite.

$$\Gamma \equiv \oint_A \vec{V} \cdot d\vec{s}$$



Circulation around a lifting airfoil.

- The lifting flow over a circular cylinder was synthesized by superimposing a uniform flow, a doublet, and a vortex.
- Recall that all three elementary flows are irrotational at all points, except for the vortex, which has infinite vorticity at the origin.
- Therefore, the lifting flow over a cylinder is irrotational at every point except at the origin.
- Similarly, the flow outside (far away from the surface) the airfoil is irrotational.
- A curve that encloses the origin, where $\nabla \times \vec{V}$ is infinite.
- If we choose a curve that encloses the airfoil, yields a finite value of Γ , equal to the sum of the vortex strengths distributed on or inside the airfoil.

Keep in mind:

- Circulation does not make lift, it just indicates it exists (footprint of the lift).
- Lift/Drag are caused by pressure and shear stress.

To get the lift:

1. Measure directly using force balancer in the wind tunnel.
2. Integrate pressure (not recommended).
3. Get the value of circulation from the velocity field (measure the flow field around the foil and calculate the circulation to get the lift force from the flow field).

Reference:

Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.